Pretrained Representations:
BART & T5

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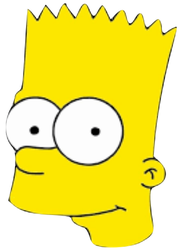
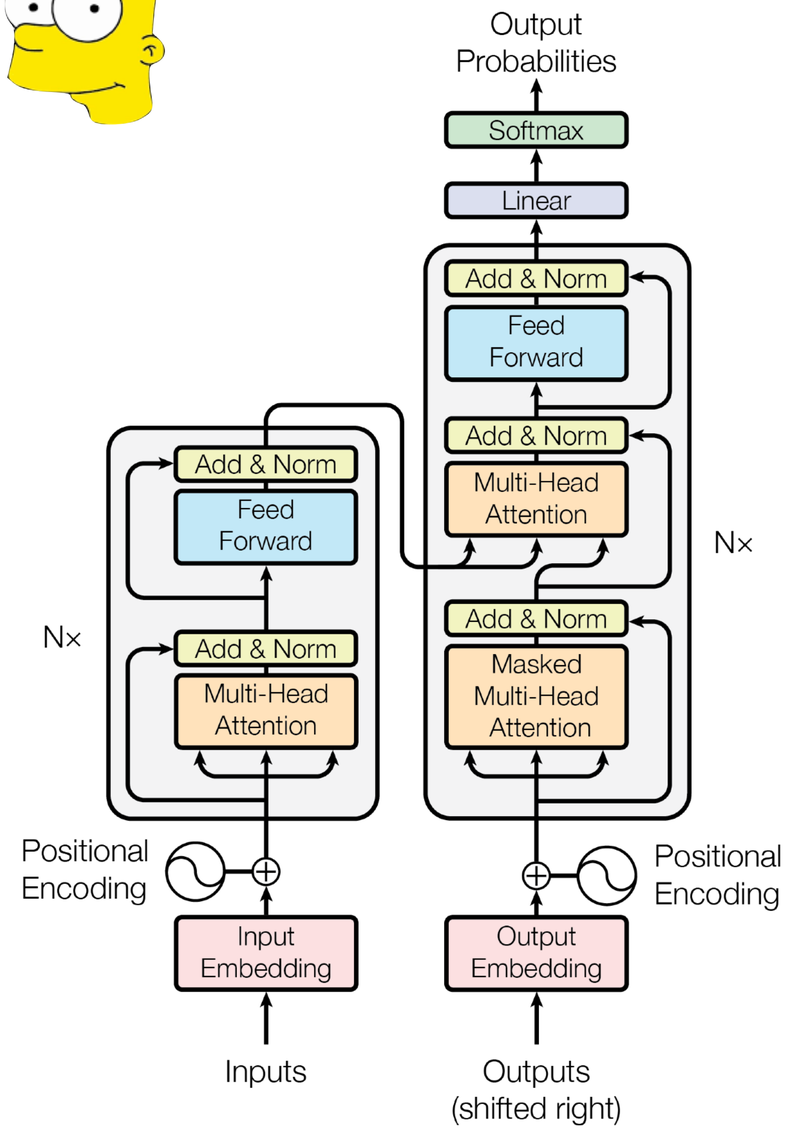
How should we pretrain sequence-to-sequence models?

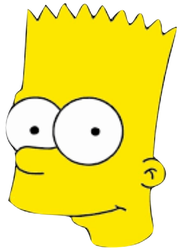
BART



- Classic transformer architecture
- Bidirectional encoder feeds into autoregressive decoder
- Cross-attention layers in decoder are back!
- **BART-base**: 6-layers each in encoder and decoder; 140M parameters
- **BART-large**: 12 layers each in encoder and decoder; 400M parameters

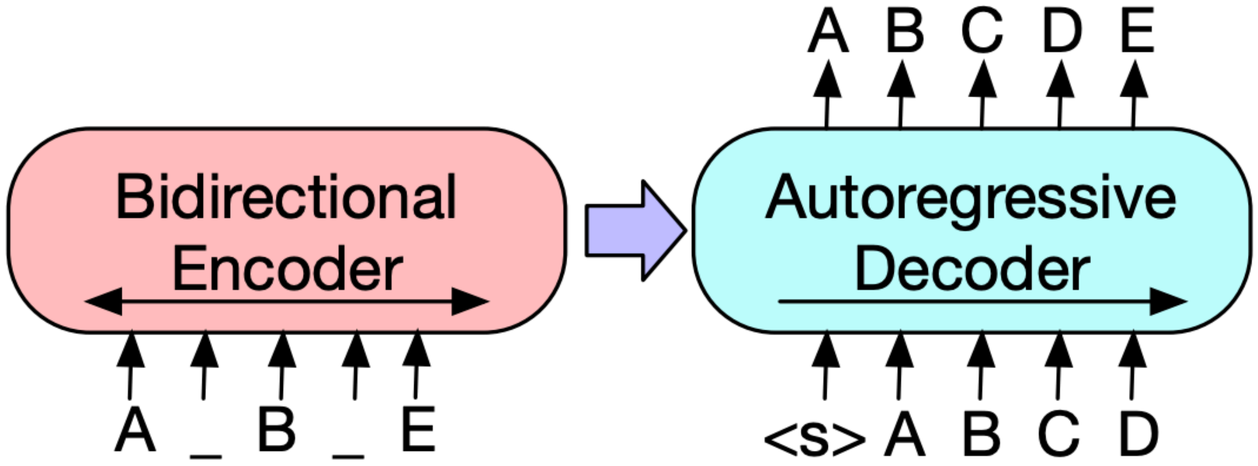


Lewis et al. (2019)

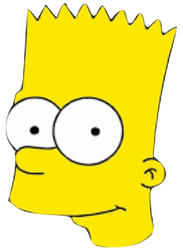


BART Pretraining

- Pretraining BART combines elements of BERT and GPT!
- **BERT-style:** input texts corrupted before they are passed to bidirectional encoder
- **GPT-style:** model is trained with a language modelling objective in the decoder: predict the next word!



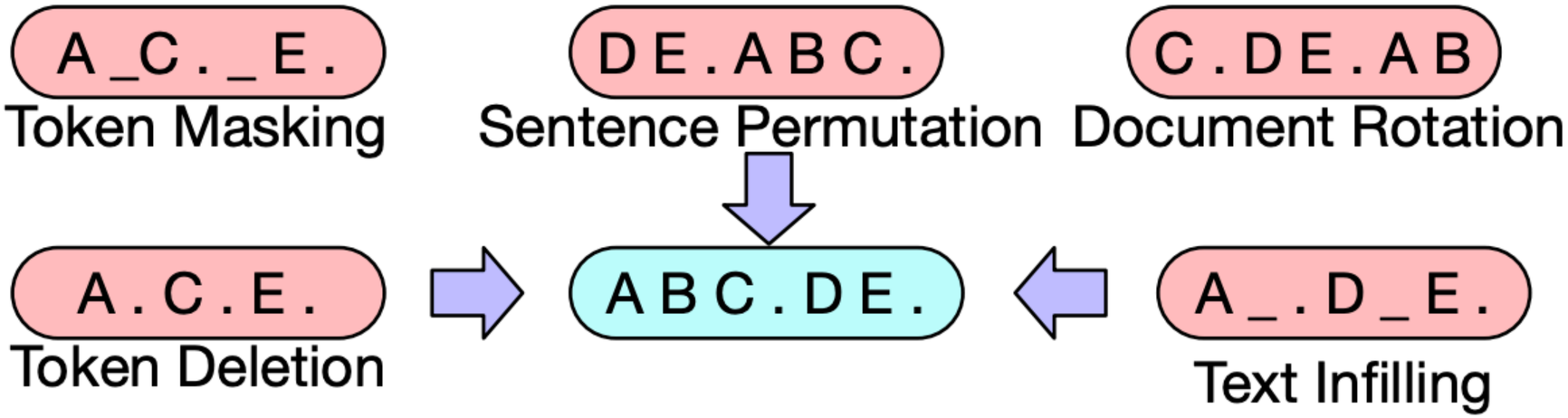
Lewis et al. (2019)



BART Pretraining

- We're not reconstructing the input the same way as BERT, so can we corrupt the input in different ways?

- Many corruption strategies can be used on the encoder side



Lewis et al. (2019)

Can do all the same tasks

- BART can also do all the tasks that BERT does!
- **Classification:**
  - Give input to both encoder AND decoder (input the sequence twice)
  - Append [CLS] token to decoder sequence and classify its output
- **Sequence Labeling:**
  - Give input to both encoder AND decoder (input the sequence twice)
  - Classify decoder output representations for each token

Can do all the same tasks

	SQuAD 1.1 EM/F1	SQuAD 2.0 EM/F1	MNLI m/mm	SST Acc	QQP Acc	QNLI Acc	STS-B Acc	RTE Acc	MRPC Acc	CoLA Mcc
BERT	84.1/90.9	79.0/81.8	86.6/-	93.2	91.3	92.3	90.0	70.4	88.0	60.6
UniLM	-/-	80.5/83.4	87.0/85.9	94.5	-	92.7	-	70.9	-	61.1
XLNet	**89.0**/94.5	86.1/88.8	89.8/-	95.6	91.8	93.9	91.8	83.8	89.2	63.6
RoBERTa	88.9/**94.6**	**86.5/89.4**	**90.2/90.2**	96.4	92.2	94.7	**92.4**	86.6	**90.9**	**68.0**
BART	88.8/**94.6**	86.1/89.2	89.9/90.1	**96.6**	**92.5**	**94.9**	91.2	**87.0**	90.4	62.8

Almost as good as RoBERTa

Way better than BERT! Why ?

Trained on way more data!

Results: Summarization

This is the first time anyone has been recorded to run a full marathon of 42.195 kilometers (approximately 26 miles) under this pursued landmark time. It was not, however, an officially sanctioned world record, as it was not an "open race" of the IAAF. His time was 1 hour 59 minutes 40.2 seconds. Kipchoge ran in Vienna, Austria. It was an event specifically designed to help Kipchoge break the two hour barrier.	Kenyan runner Eliud Kipchoge has run a marathon in less than two hours.
PG&E stated it scheduled the blackouts in response to forecasts for high winds amid dry conditions. The aim is to reduce the risk of wildfires. Nearly 800 thousand customers were scheduled to be affected by the shutoffs which were expected to last through at least midday tomorrow.	Power has been turned off to millions of customers in California as part of a power shutoff plan.

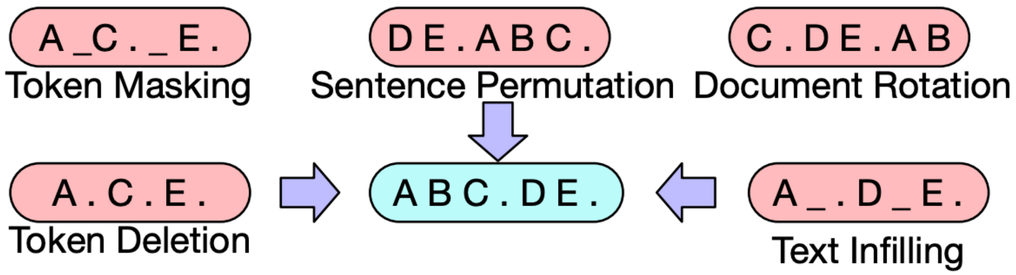
However, BART can do generation tasks too

Decoder is autoregressive!

Lewis et al. (2019)

Which encoder-side corruption?

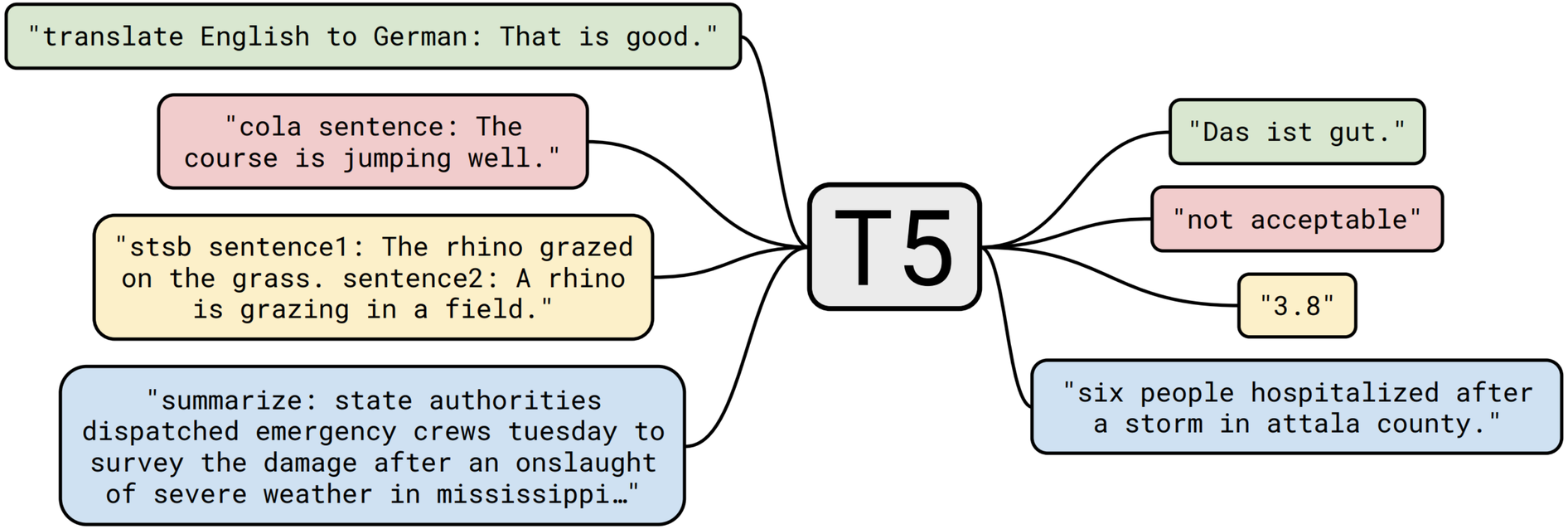
Model	SQuAD 1.1 F1	MNLI Acc	ELI5 PPL	XSum PPL	ConvAI2 PPL	CNN/DM PPL
BART Base						
w/ Token Masking	90.4	84.1	25.05	7.08	11.73	6.10
w/ Token Deletion	90.4	84.1	24.61	6.90	11.46	5.87
w/ Text Infilling	**90.8**	84.0	24.26	**6.61**	**11.05**	5.83
w/ Document Rotation	77.2	75.3	53.69	17.14	19.87	10.59
w/ Sentence Shuffling	85.4	81.5	41.87	10.93	16.67	7.89
w/ Text Infilling + Sentence Shuffling	**90.8**	83.8	24.17	6.62	11.12	**5.41**



- Different corruption better for transfer to different tasks
- **Use combination of text infilling + sentence permutation**

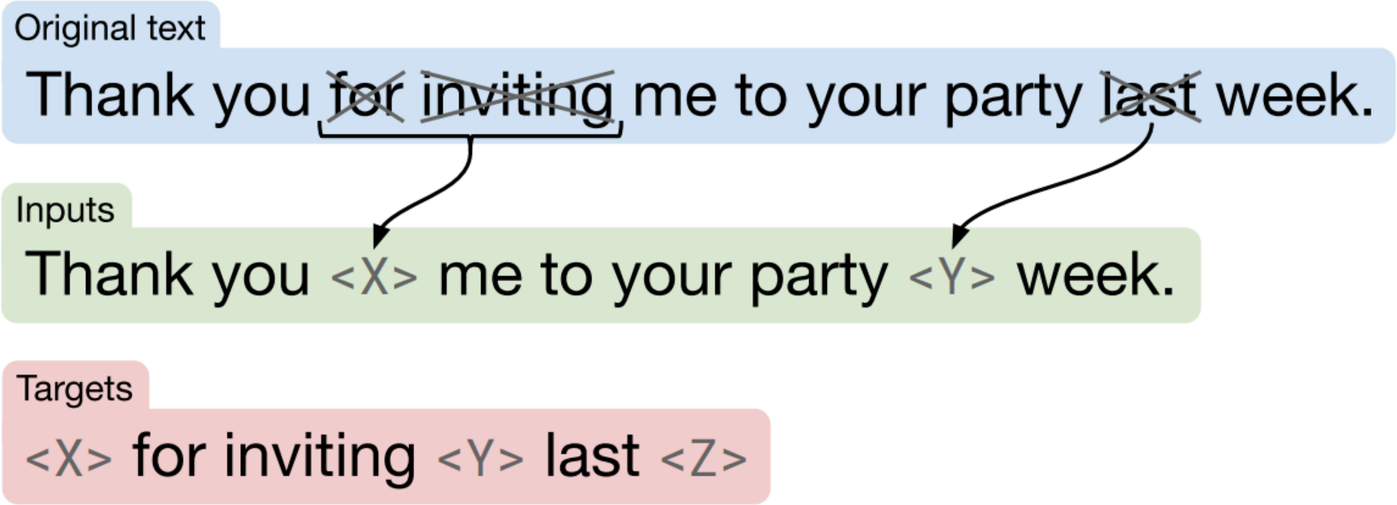
T5

- **Similar idea as BART:** Any problem can be cast as sequence-to-sequence



Raffel et al. (2019)

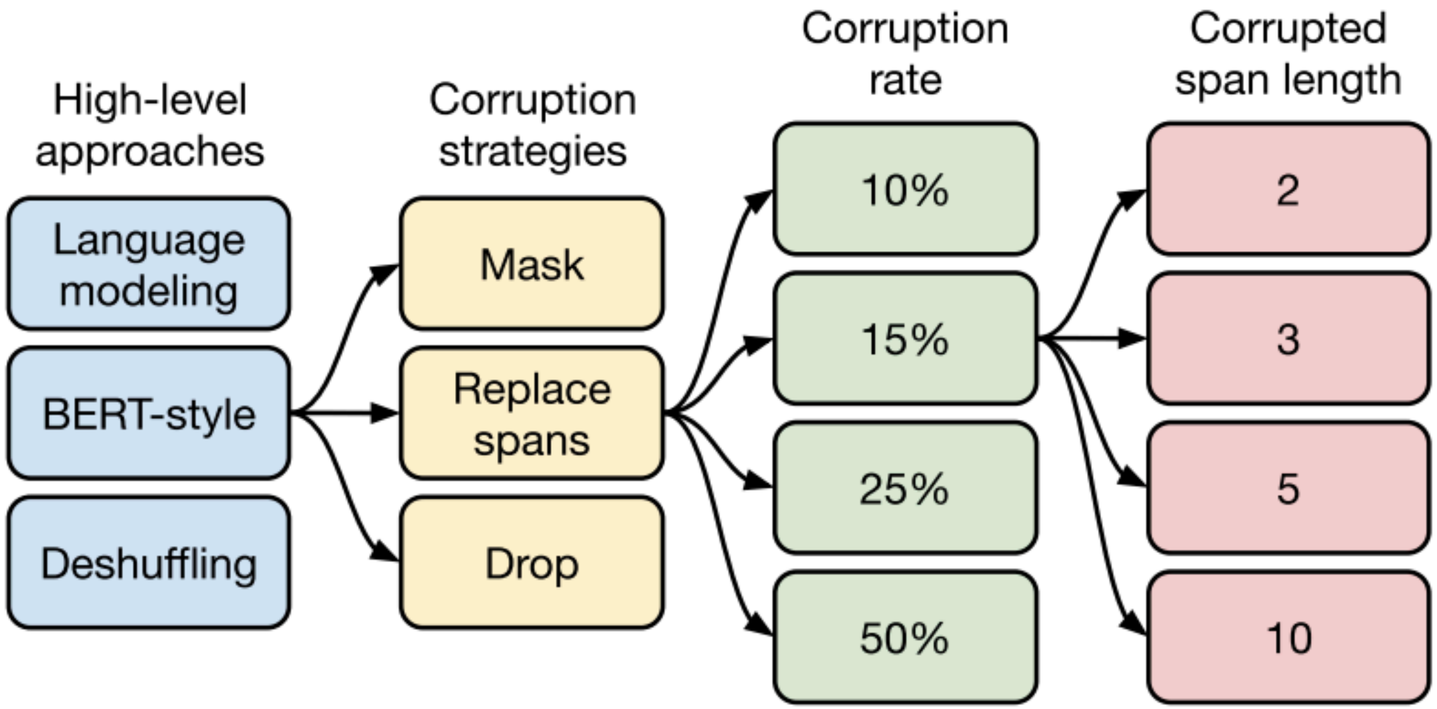
T5 Pretraining



- Similar to BART
- Uses the infilling objective where tokens are reconstructed from underspecified mask corruptions

Raffel et al. (2019)

T5 Pretraining Decisions



- Explored many dimensions of pretraining in se2seq framework
- Took findings to train much larger model — 11B parameters!

Raffel et al. (2019)

Recap

- **Contextual representations:** Let us model words and sequences conditioned on the context around them
- **ELMo:** Based on bidirectional LSTMs. **Good for pretrained embeddings.**
- **GPT**: Uses a transformer decoder. **Good for generating text as a language model.**
- **BERT**: Uses a transformer encoder. **Good for classification and sequence labelling.**
- **BART + T5:** Pretraining sequence-to-sequence transformer models. **Extendable to all task types!**

References

- Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., ... & Zettlemoyer, L. (2020). BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. *ACL*.
- Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., ... & Liu, P. J. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. *JMLR*.